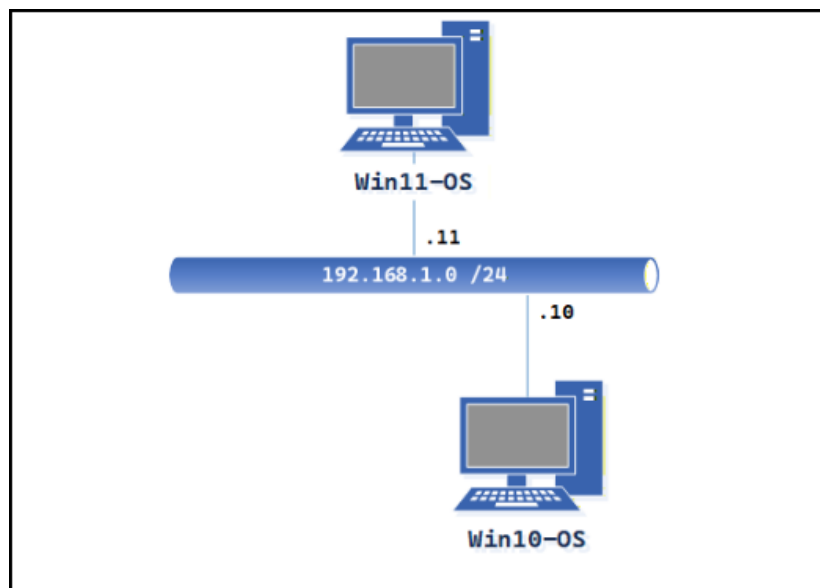




A+ LAB SERIES V5

Lab 0: Explore the Netlab Environment



Objective

In this lab, you will examine the lab environment and the appropriate required components to complete the labs in the course. Additionally, you will have the opportunity to review the different steps to complete the labs in a reasonable time frame and to achieve positive outcomes in the lab environment.

Lab Settings

The information on the table below will be the consistent password scheme for the bulk of the labs.

Virtual Machine	IP Address	Account (if needed)	Password (if needed)
Win10-OS	192.168.1.10 / 24	sysadmin	NDGLabpass123!
Win11-OS	192.168.1.11 / 24	sysadmin	NDGLabpass123!

Objectives

Part 1: Explore the Netlab Environment

Part 2: Access the Cybersecurity Essentials Lab VM

Introduction

Proper utilization of the Netlab virtualized environment is important to your understanding and positive learning outcome in this class. To assist students in exploring and evaluating the lab environment this lab will provide direct access to the environment. Once you have accessed the environment you will explore the interface and discover the proper utilization of the Netlab resource. Students will then create reservations on their own to complete labs.

Required Resources

PC or Mac with internet access

Background / Scenario

The practical component of this class is Netlab. You will use this resource to complete the lab activities in NetAcad that require computing resources. These labs are written specifically for Netlab. Some of the labs will be written and intended to be completed in Oracle VirtualBox in the Netlab environment. We will complete all of these in a managed environment in NetLab. This will prevent you from having to install software on your own computer. Some software can create compatibility issues on your computer, and we would like to avoid this. Simply download the lab worksheet from within Brightspace and read over it before beginning. Select the desired lab you are required to complete, and you will be automatically provided a link into the lab environment. You will create a reservation and access the reservation from within the reservation manager. Once you log in you will be placed onto the VM host computer. You will use the specific host machine as a desktop to complete the labs. All students must complete the labs in the Netlab environment. Any screenshots must be submitted to the lab exams to count towards credit.

These labs will require two resources: the NDG lab environment and the NDG Lab Worksheets to complete the lab. As you complete the lab you will take screenshots and the screenshots will be uploaded to the graded Lab quiz link according to the instructions below. The Netlab resources compose a practical learning platform, and the following course mechanisms are described in length below:

Netlab Student Lab Links

Each link is located in the **Content > Coursework > Netlab Student Labs folder**. Once you review which lab you have been assigned to Netlab, you will return to Brightspace and locate the lab in the Netlab Student

Labs folder. The lab link requires a reservation, and you must complete the lab during that attempt. There are limited lab resources assigned for each lab, so the reservations can be full, but this is unlikely in this course.

Lab Worksheets

These lab documents are located in the **Content > Coursework > Netlab Student Worksheets** folder. Each lab has a specific worksheet to complete. You must complete these labs in the Netlab environment. When finished the specific screenshots are submitted in the appropriate lab quiz in Brightspace.

Required Screenshot List

The required list of screenshots (there will always be four) are listed in each lab's Upload Link. Refer to this first and write these down. Then download the worksheet and review it before starting the lab. Once the lab starts you will take the screenshots as you go.

Part 1. Explore the Netlab Environment

As you are completing the activities in each unit you will routinely come across two types of lab activities: A classroom lab or a practical lab. The classroom labs simply require you to research on your own or they are group activities that we will complete in class. The research activity labs can be performed with research on the Internet. But the classroom labs will be administered by the instructor. These labs are completed and are rarely submitted by you for grading. These are for you to discover or enhance the concepts you have been reviewing in the course.

The practical labs will be used to explain a concept or show you how to perform a specific task. These tasks are often used in the workplace as part of a specific task or job role. Practical labs require the use of specific lab environments. These environments leverage the use of computing resources that require virtualized operating systems and hardware. All of the lab worksheets provided by NDG for Netlab and any custom sets available in Brightspace require you to use specific hardware and software that you cannot install on your own computer. To prevent any potential adverse effects on your hardware, you have been given access to a pre-configured interface known as Netlab.

When you need to complete one of these labs in the course take note of the number of the lab. For example, some of the labs will be completed out of order so make sure you are working in the correct lab and do not waste your time. Record this number and you will then return to your Brightspace course shell. Once in Brightspace, locate the **Netlab Student Lab** folder which is located under Coursework. You should have already located this during your first week of access to the course. Locate the worksheets within the **Netlab Student Worksheets** folder and download it.

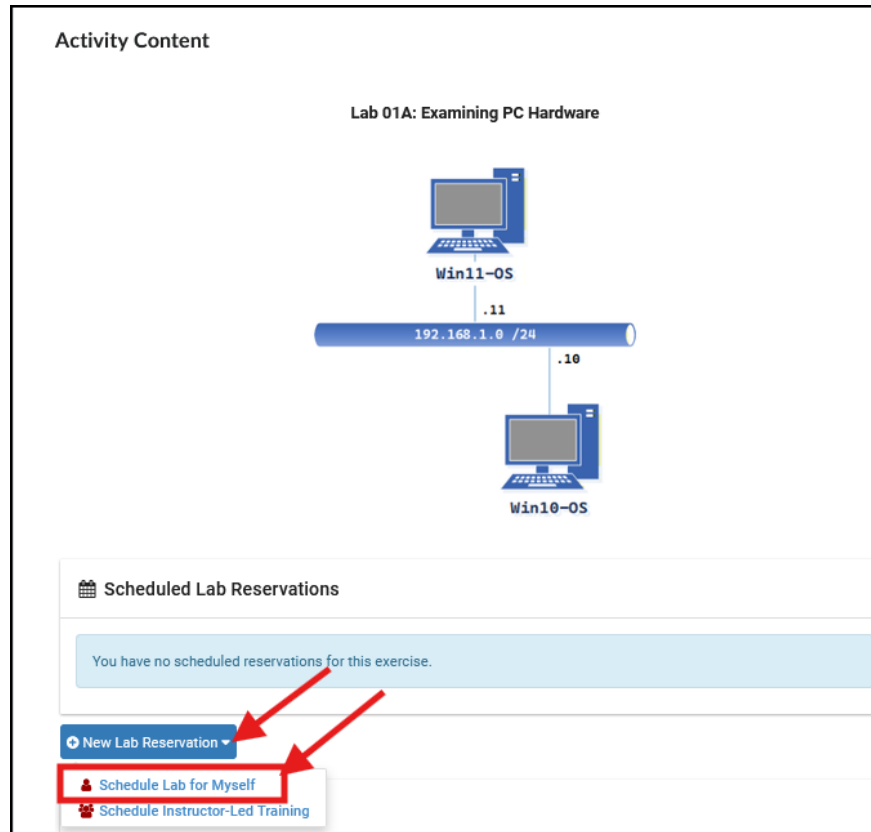
Step 1: Access the Netlab Environment

a. Locate the correct lab worksheet and download this to your local computer. Optimally you should review the worksheet and become familiar with what you will be completing within the lab environment. Locate and navigate to the appropriate unit folder and locate the Lab X Upload Link for the current lab's screenshot requirements. The Online Labs - Netlab provides a list of the Netlab labs that will be submitted for grading. Each lab upload link contains a detailed description of a list of screenshots you will be required to take while completing the lab. Typically, there are four screenshots, the first of which is always the same. The others are contingent upon what work is being completed and which steps are critical.

Critical steps are determined by the instructor, and the remaining three screenshots are required to be recorded. All of these screenshots will be submitted to the appropriate lab link in **Brightspace > Content > Coursework > Unit X > Lab X Upload Link**. Now that you know which screenshots are required and you have the worksheet you can now access the lab.

b. Click the appropriate **Netlab Student Labs** link for your lab.

c. Next in the **Activity Content** window you will locate the New lab Reservation button and click it. A context sub menu is displayed that allows for scheduling of access. Select the **Schedule Lab for Myself** option. An example of this is shown in the image below.



d. A new window will open with the Pod Scheduler availability listed. Note the month and day listed in the top left example. At the top right the current time is listed. Below is a view of **some** of the pods available to choose from. Note there is an arrow that allows you to view other available pods as well. If pods 1-4 are full, then navigate to the right or even further to scroll through the remaining available pods.

To select a pod, locate the first available pod. The current time is represented by a red line. If you choose to select the current time you must click above this red line. An example is provided in the image below.

The screenshot shows the 'Pod Scheduler' interface. At the top, there's a calendar for August 2026. A red box highlights the calendar, the 'Selected Day' (August 31, 2026), and a 'Current Time' clock showing 21:34 Eastern Time (US & Canada). Below this is a grid of reservation slots for four NDG A+ v5 Student Pods. A red box highlights a reservation for Pod 01 from 21:00 to 22:00, with a red arrow pointing to it.

e. A new window with the Add Reservation window will be displayed. If you selected the current time, the start time will be grayed out. If you are choosing a future time enter that time now.

f. Your end time should be selected based on your evaluation of the amount of time you estimate you will need. Typically, your labs should take no more than 1 hour on the first attempt. Subsequent attempts will become shorter as your skills improve. Note the total length of your reservation and then click the **Submit** button when you are ready. Note the example below.

Note: The last 15 minutes of your reservation will be reserved for resetting the VM to its original state. Therefore, you will not have access to the entire time. Plan accordingly.

The screenshot shows the 'Add Reservation' window. It contains the following fields: Pod (NDG A+ v5 Student Pod 01), Reservation Type (Instructor Private Reservation), Reserve For (Mark Turner), Lab Exercise (Lab 01A: Examining PC Hardware), Time Zone (Eastern Time (US & Canada)), Start Time (2026-08-31 21:35), End Time (2026-08-31 22:30), and Length of Reservation (44 mins.). A red box highlights the Start Time and End Time fields. At the bottom, there are buttons for 'Submit', 'Previous', and 'Cancel'. A red arrow points to the 'Submit' button.

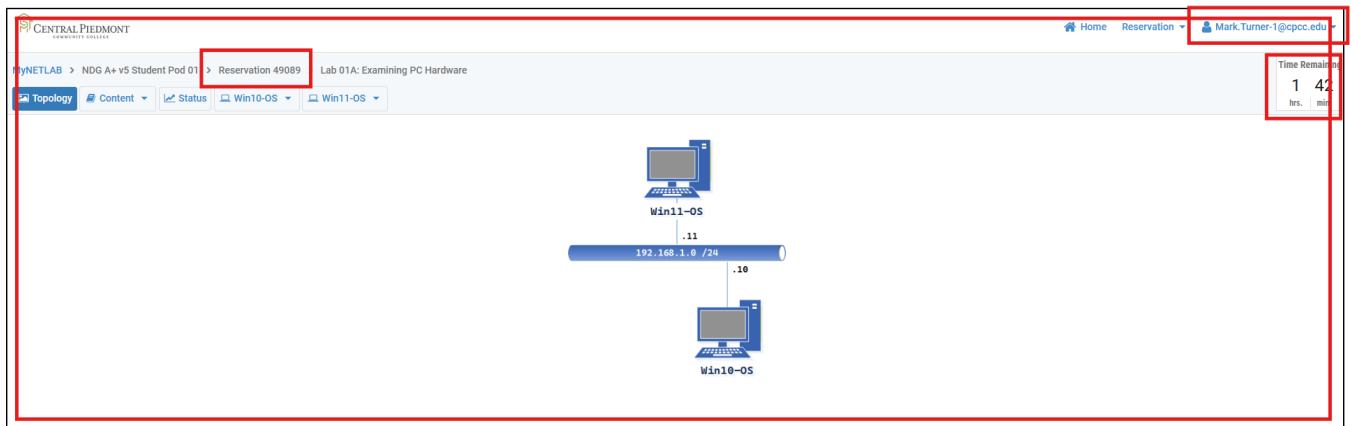
g. Click **OK** on the Reservation Scheduled window.

Lab - Explore the Netlab Environment

h. Note your reservation is now populated with all relevant details in the Lab Reservations window. You may make more reservations if you wish. Please do not make reservations that overlap as this is not fair to other students. Click the **Enter Lab** button when you are ready to enter.

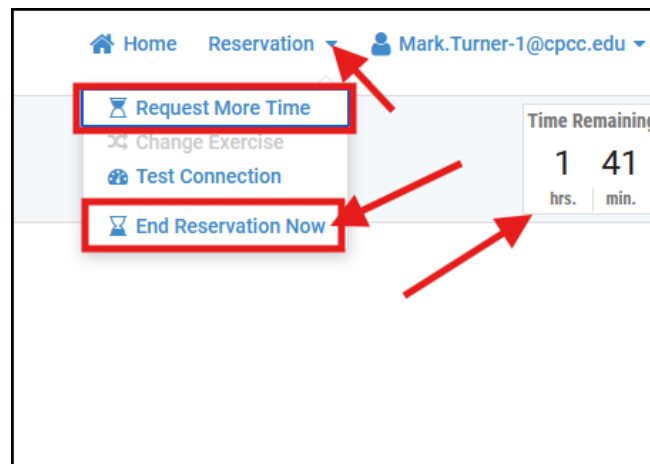
Step 2: Explore the Netlab Environment

a. After the lab environment loads you will be presented with a topology. This is where your first lab screenshot is always generated. Using the snipping tool/screenshot utility to create the screenshot now. Verify that the entire window is captured. For a valid example, review the screenshot below. It should contain your login on the top right and the reservation on the left with the topology visible.



b. Click on **Reservation** in the top right. A context menu will provide you further options. Not all labs allow you to request more time. If they are available then there is a limited number that will only allow 30 minutes each additional attempt. You can do this using the **Request More Time** option. Note the time at the right. It is your responsibility to manage your time as once this runs out, your access will be removed, and you will lose all progress. When you have completed all components of the lab you will use the **End Reservation Now** function.

Note: Once a reservation ends all the pod and all hosts are reset back to their original state. Any information or progress will be erased and is irrecoverable. Please monitor this time as students regularly struggle with this point, and it can be disheartening to see a student lose progress over this. You can attempt all labs as many times as you wish before the due date. This is contingent on lab resources being available and pods open for reservations.



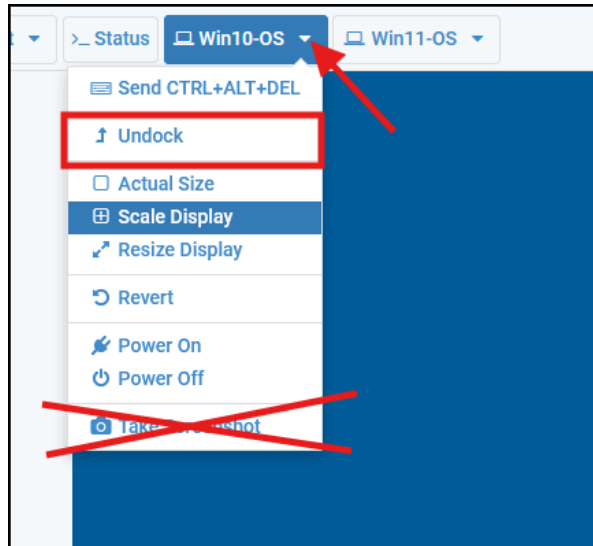
c. Click on the **Content** tab for a copy of the worksheet that is used for the lab.

d. Click on **Status** to view the status of the lab environment. This provides you with the current state of the lab hosts and equipment.

Lab - Explore the Netlab Environment

e. Click on the **Windows Host** tab to access Lab VM. There may be more than one host to review in the lab. A context menu will appear with options. You can undock the VM window for easy access and if you have multiple monitors. This is a helpful option if you want to access two different hosts (if available).

Do not use the Take Screenshot option. These screenshots are retained in the VM and will be lost once the reservation is over. This is the incorrect method, and the instructor has no way to view these.



f. Once you have access to the host use the password for the “**sysadmin**” account to access the environment. The password is “**ndglabpass123!**”.

g. Note the desktop of the host is available and displays the desktop background. Any instructions in the lab worksheets that ask you to complete a task will reference a specific VM name for each component of the lab. Follow the instructions and practice interacting with the environment. If you have questions feel free to reach out to your instructor.